INCH-POUND

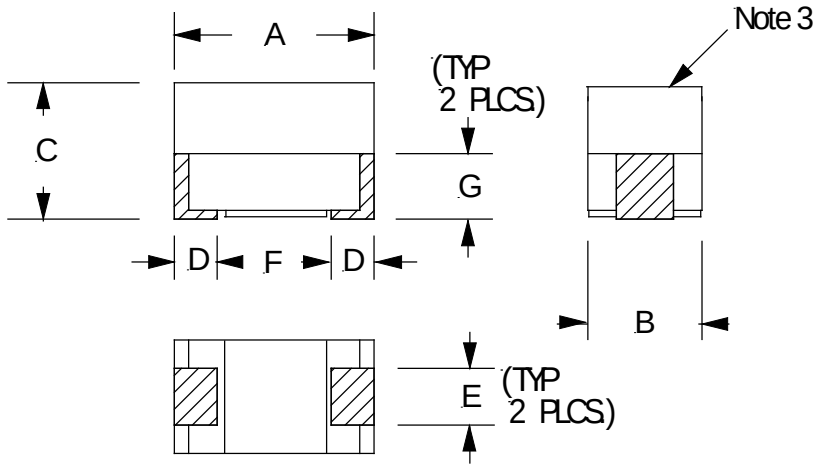
MIL-PRF-32560/3
25 October 2016

PERFORMANCE SPECIFICATION SHEET

COIL, RADIO FREQUENCY, CHIP, FIXED, MOLDED, UNSHIELDED
ESTABLISHED RELIABILITY, SURFACE MOUNT

This specification is approved for use by all Departments
and Agencies of the Department of Defense.

The requirements for acquiring the products described
herein shall consist of this specification sheet and MIL-PRF-32560.



Dim.	Inches	Millimeters
A	0.118 to 0.138	3.00 to 3.51
B	0.085 to 0.105	2.16 to 2.67
C	0.081 to 0.101	2.06 to 2.57
D	0.016 min.	0.41 min.
E	0.041 to 0.061	1.04 to 1.55
F	0.070 (Ref. Only)	1.78 (Ref. Only)
G	0.054 (Ref. Only)	1.37 (Ref. Only)

NOTES:
1. Dimensions are in inches.
2. Metric equivalents are given for general information only.
3. Marking shall be on the top surface of the coil.

FIGURE 1. Dimensions and Configuration.

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REQUIREMENTS:

Dimensions and configuration: See Figure 1.

Material: Phenolic core (inductance values 0.0018 to 0.082 uH, inclusive); Iron core (inductance values 0.10 to 0.82 uH, inclusive); Ferrite core (inductance values 1.0 to 100 uH, inclusive).

Termination finish: Tin lead only

Weight: 0.10 gram, maximum.

Operating temperature range: -55ºC to +125ºC.

Temperature rise (at 90ºC): 35ºC maximum. Test performed with coil mounted on SMD test substrate.

Maximum operating temperature: +125ºC.

Altitude: 70,000 feet maximum.

Dielectric Withstanding Voltage: In accordance with MIL-PRF-32560.

Barometric Pressure: In accordance with MIL-PRF-32560.

Insulation Resistance: In accordance with MIL-PRF-32560.

Points of test voltage application for Dielectric Withstanding Voltage, Barometric Pressure, and Insulation Resistance are between the terminals of the coil shorted together and the contact arm or assembly. The contact arm is made of conductive, moisture-resistant, resilient material and it shall cover the entire surface opposite the terminals, and shall be held firmly in place against this surface during testing.

Electrical characteristics (initial): See Table I.
Inductance and tolerance: See Table I.
Quality Factor (Q): See Table I.
Self-resonant frequency: See Table I.
DC resistance: See Table I.

Terminal (Bond) Strength: 2.0 pounds, when tested in accordance with MIL-STD-883, Method 2011, Test condition F. Test performed with coil mounted on SMD test substrate.

Solderability: In accordance with MIL-PRF-32560 with the following exception:
a. MIL-STD-202-208, test method B

Resistance to solvents: Test is not applicable.

Overload: Test performed with coil mounted on SMD test substrate.

Low temperature storage: Test performed with coil mounted on SMD test substrate.

Vibration: In accordance with MIL-PRF-32560. Test performed with coil mounted on SMD test substrate.

Shock (Specified Pulse): In accordance with MIL-PRF-32560. Test performed with coil mounted on SMD test substrate.

Life: Test performed with coil mounted on SMD test substrate.

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Moisture Resistance: In accordance with MIL-PRF-32560.

SMD test substrate (where applicable): Material shall be made of a minimum of 96 percent alumina, or equivalent. The test substrate shall be prepared with metallized surface land areas.

Coil Mounting (where applicable): Test coils are to be soldered to the SMD test substrate's metallized areas only. Solder used shall be Sn63Pb37 solder or equivalent. The test substrate shall then be placed in or on a suitable heat transfer unit (molten solder, hot plate, tunnel oven, etc.) with the temperature maintained at 260ºC +/- 5ºC, until the solder melts and reflows forming a homogenous solder connection.

Electrical characteristics (final): See Table II. For any Subgroup test requiring coils to be mounted to an SMD test substrate, the electrical characteristics (final) measurements are to be referenced to the electrical characteristics (initial) measurements determined after the test coil is mounted to the test substrate.

Part Marking: Coil marking is not applicable due to body size constraints.

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Table I Electrical characteristics (initial) and dash numbers.

Dash Number 1/	Inductance (µH) 3/	Inductance Tolerance (%)	Q (min) 3/	Test Frequency (MHz)	Self-resonant frequency (MHz) (min)	DC resistance (ohms) (max)	Current rating (mA) (max) 2/
BR0018*B*	0.0018	±20	40	50	2700	0.050	1,562
BR0022*B*	0.0022	±20	40	50	2700	0.050	1,562
BR0027*B*	0.0027	±20	40	50	2700	0.050	1,562
BR0033*B*	0.0033	±20	40	50	2700	0.050	1,562
BR0039*B*	0.0039	±20	40	50	2700	0.050	1,562
BR0047*B*	0.0047	±20	40	50	2700	0.050	1,562
BR0056*B*	0.0056	±20	40	50	2700	0.050	1,562
BR0068*B*	0.0068	±20	40	50	2700	0.050	1,562
BR0082*B*	0.0082	±20	40	50	2700	0.050	1,562
BR010*B*	0.010	±10	30	50	2000	0.13	966
BR012*B*	0.012	±10	30	50	1850	0.14	931
BR015*B*	0.015	±10	30	50	1700	0.16	871
BR018*B*	0.018	±10	30	50	1550	0.18	821
BR022*B*	0.022	±10	30	50	1300	0.20	779
BR027*B*	0.027	±10	30	50	1150	0.22	743
BR033*B*	0.033	±10	30	50	1000	0.24	711
BR039*B*	0.039	±10	30	50	900	0.27	670
BR047*B*	0.047	±10	30	50	800	0.30	636
BR056*B*	0.056	±10	30	50	750	0.33	606
BR068*B*	0.068	±10	30	50	700	0.36	580
BR082*B*	0.082	±10	30	50	625	0.40	551
BR10*B*	0.10	±10	30	25	550	0.20	1,018
BR12*B*	0.12	±10	30	25	500	0.22	971
BR15*B*	0.15	±10	30	25	450	0.25	910
BR18*B*	0.18	±10	30	25	400	0.28	860
BR22*B*	0.22	±10	30	25	350	0.32	805
BR27*B*	0.27	±10	30	25	320	0.36	759
BR33*B*	0.33	±10	30	25	300	0.40	720
BR39*B*	0.39	±10	30	25	250	0.45	679
BR47*B*	0.47	±10	30	25	220	0.50	644
BR56*B*	0.56	±10	30	25	180	0.55	614
BR68*B*	0.68	±10	30	25	160	0.60	588
BR82*B*	0.82	±10	30	25	140	0.67	556
B1R0*B*	1.0	±5	30	7.9	120	0.70	515
B1R2*B*	1.2	±5	30	7.9	100	0.75	497
B1R5*B*	1.5	±5	30	7.9	85	0.85	467
B1R8*B*	1.8	±5	30	7.9	75	0.90	454
B2R2*B*	2.2	±5	30	7.9	70	1.0	431
B2R7*B*	2.7	±5	30	7.9	65	1.1	411
B3R3*B*	3.3	±5	30	7.9	58	1.2	393
B3R9*B*	3.9	±5	30	7.9	50	1.3	378
B4R7*B*	4.7	±5	30	7.9	45	1.5	352
B5R6*B*	5.6	±5	30	7.9	42	1.6	341
B6R8*B*	6.8	±5	30	7.9	40	1.8	321
B8R2*B*	8.2	±5	30	7.9	35	2.0	305
B100*B*	10	±5	30	7.9	30	2.1	297

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Table I Electrical characteristics (initial) and dash numbers. - continued

Dash Number 1/	Inductance (µH) 3/	Inductance Tolerance (%)	Q (min) 3/	Test Frequency (MHz)	Self-resonant frequency (MHz) (min)	DC resistance (ohms) (max)	Current rating (mA) (max) 2/
B120*B*	12	±5	30	2.5	28	2.5	272
B150*B*	15	±5	30	2.5	25	2.8	257
B180*B*	18	±5	30	2.5	23	3.3	237
B220*B*	22	±5	30	2.5	20	3.7	224
B270*B*	27	±5	30	2.5	18	5.0	193
B330*B*	33	±5	30	2.5	15	5.6	182
B390*B*	39	±5	30	2.5	14	6.4	170
B470*B*	47	±5	30	2.5	13	7.0	163
B560*B*	56	±5	30	2.5	11	8.0	152
B680*B*	68	±5	30	2.5	10	9.0	144
B820*B*	82	±5	30	2.5	9	10.0	136
B101*B*	100	±5	30	2.5	8	13.0	120

1/ The complete dash number will include three (3) additional letters (indicated by ***). The first additional letter will indicate the inductance tolerance (e.g. J = ±5%, K = ±10%, M = ±20%), the second additional letter will indicate the termination finish B = Tin Lead, and the third additional letter will indicate the product level (e.g. M, P, R, S) and will be added to the end of the dash number.
2/ Maximum current allowed not to exceed the specified maximum temperature rise.
3/ Inductance and Q to be measured using HP4291A; HP 4194A or equivalent using TF16092; TF16034 or equivalent.

Table II Electrical characteristics (final).

Inspection group	Allowable variation from initial measurements			
	Inductance (Percent)	DC resistance	Self-resonant frequency (Percent)	Q (Percent)
Qualification inspection				
Group II	± 5	±(3% +.001 ohm)	-10	-10
Group IIIA	± 5	-------	----	-10
Group IV	± 5	±(3% +.001 ohm)	-10	-10
Group V	± 5	±(3% +.001 ohm)	-10	-10
Group B inspection				
Subgroup 1	± 5	±(3% +.001 ohm)	-10	-10
Subgroup 2	± 5	±(3% +.001 ohm)	-10	-10
Subgroup 3a	± 5	----	----	-10
Subgroup 4	± 5	±(3% +.001 ohm)	-10	-10

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Part or Identifying Number (PIN): The part number shall be in the following form.

M32560/03-******* (dash number per Table I)

Referenced documents: In addition to MIL-PRF-32560, this document references:

MIL-STD-202-208, MIL-STD-883

Custodians:
Army – CR
Navy – EC
Air Force – 85
DLA – CC

Preparing activity:
DLA – CC

Review Activities:
Army – AR, CR4, MI
Navy – AS, CG, MC, OS
Air Force – 19, 99
NASA – NA
Other - MDA

(Project 5950-2017-003)

NOTE: The activities listed above were interested in this document as of the date of this document. Since organizations and responsibilities can change, you should verify the currency of the information above using the ASSIST Online database at https://assist.dla.mil.

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